Final Report Results

The revised fact checker and the final reruns of the 18 trials: what changes in the report, paste-ready text for Methods, Results, Discussion and Limitations, and Tables 1 to 8. Dialect labels were used only in the evaluation.

0.69 → 0.80
checker accuracy on the held-out topics (McNemar p = 0.0125)

0 of 210
draft claims contradicted by their tweets

0 of 18
summaries changed by the agent

0.0024
mean Equal Coverage in every condition

What changes in the report

> **Correction to the poster.** The poster's "CONTRADICTED 3 → 0" is not an improvement by the agent. All three contradictions were checker errors (information was missing; no tweet said the opposite). Present it as a finding of the checker validation, as in the Results text below.

1. Replace the poster's fact and fairness numbers with the ones on this page.
2. Add the checker validation to Methods and Results (text and Tables 1 to 3).
3. Report the final runs with Tables 4 to 8.
4. Do not report the "summaries missing a viewpoint" count printed in `evaluation.log`; that measure is flawed.
5. Two team members label 60 claims (instructions at the end) so we can fill in the `[κ]` placeholders.

Methods

Validation of the fact checker

The fact checker produces every factual verdict we report and is the only trigger for a revision (a CONTRADICTED claim), so we validated it before the final runs. In our first label-free run, 161 of the 210 claims extracted from the 18 drafts (77%) were judged UNKNOWN. To measure how many verdicts were wrong, we labelled all 210 claims as SUPPORTED, CONTRADICTED or UNKNOWN against the tweets each claim cites, following a written guideline. Support may be paraphrased. A claim with "some", "others" or a plural needs one supporting tweet, and a claim with "several", "many", "frequently" or "often" needs three. A share of the whole collection ("most", "a significant portion") cannot be established from cited tweets. Every detail in a list needs support, other users' opinions never contradict "some users say X", and missing content is never a contradiction. The reference labels were produced by an LLM (Claude), independently of Qwen's verdicts. Two team members labelled a random sample of 60 claims independently; their agreement with the reference was `κ = [..]` and `κ = [..]`, and with each other `κ = [..]`.

We made every design decision on the development topic, McDonalds (78 claims), and evaluated the final design once on the held-out topics, Obama and Superbowl (132 claims). Every verdict was a live call to Qwen3.5-4B with the production settings (temperature 0, seed 42). We report accuracy, the precision and recall of SUPPORTED, false UNKNOWN (a supported claim left unconfirmed), false SUPPORTED and false CONTRADICTED, and compare the original and revised checkers on the same claims with an exact McNemar test.

The revised checker

The revised checker differs from the original in five ways. (1) Its instructions state the counting and attribution rules above. (2) It writes its explanation and the quoted references before the verdict: with JSON-schema constrained decoding the fields are generated in order, so the model checks each detail of the claim before it decides. (3) It shows every cited tweet before any extracted interpretation, up to 40 evidence items; the original showed at most 16 items ranked by word overlap, which hid the supporting tweets of claims citing ten or more tweets. (4) Requests may be up to 40 KB and answers up to 1,500 tokens, instead of 16 KB and 900 tokens, limits that had made 13 checks fail. (5) Two rules are applied in code after the model's verdict. A claim about a share of the whole collection is never SUPPORTED, and a claim with "many", "several" or "frequently" is SUPPORTED only when its references quote three distinct tweets. A claim about what users say is never CONTRADICTED, since other users saying something else does not refute it.

Final runs

With the revised checker we reran all 18 trials (3 topics × 2 prompts × 3 seeds). The drafts and their extracted claims were reused from the earlier run and every claim was checked anew, so the runs differ only in the checker and the feedback layer. A second run added the proportion feedback layer: Qwen reads all tweets of the topic and the summary and judges whether the summary is in proportion to the tweets. When it names an under-represented viewpoint that no verified claim already reports, the summary is revised to cover it, and the revision is kept only if the checker supports it. In both runs the agent never saw a dialect label: the labels were removed from its graph and every model request was checked for them. The labels were used only in the evaluation.

Fairness metrics

For each dialect group (AA, Hispanic, White), group coverage is the mean entailment probability from roberta-large-mnli over all pairs of a tweet of that group (premise) and a summary sentence (hypothesis). Equal Coverage (EC) is the mean absolute deviation of the three group coverages from the overall coverage of all pairs: 0 means the groups are covered equally, and lower is fairer. We also report the coverage of the worst-covered group and the four-fifths disparity, 1 − (lowest group coverage / highest group coverage); a value of at most 0.20 passes the four-fifths rule. Differences between the groups over the 18 drafts were tested with a Friedman test and paired Wilcoxon signed-rank tests.

Results: the fact checker

The original checker

Against the reference labels, the original checker reached an accuracy of 0.67 on the 210 claims, with a precision of 0.78 and a recall of 0.38 for SUPPORTED (Table 1). It was mainly too strict: 57 supported claims were left UNKNOWN, against 10 unsupported claims judged SUPPORTED. All three CONTRADICTED verdicts were wrong; in each, information was missing and no tweet stated the opposite. Thirteen checks never reached a verdict because the request exceeded 16 KB or the answer hit the 900-token limit. Half of the claims the reference leaves UNKNOWN (58 of 116) are supported by an uncited tweet of the same topic: the summary cited the wrong tweets, which no checker that reads only the cited tweets can confirm. Qwen was also not reproducible across sessions: byte-identical requests returned a different verdict in 2 of 5 cases, despite temperature 0 and a fixed seed.

Table 1. Original checker against the reference labels, all 210 claims.

Reference label	Checker: SUPPORTED	CONTRADICTED	UNKNOWN	Total
SUPPORTED	36	1	57	94
UNKNOWN	10	2	104	116
Total	**46**	**3**	**161**	**210**

Accuracy 0.67; SUPPORTED precision 0.78, recall 0.38. 13 checks failed (input or answer limit). 58 of the 116 claims labelled UNKNOWN are supported by an uncited tweet of the topic. The reference contains no CONTRADICTED claim.

Designs on the development topic

Table 2 compares the designs on the development topic. Explicit rules with higher limits (A) raised recall from 0.38 to 0.78 but accepted lists with an unsupported item (precision 0.64). Writing the explanation before the verdict (B) restored precision to 0.82. Letting the model list the missing details and applying the counting rules in code (D) failed: the 4B model rarely left the list empty and filled it with words of the claim itself. Showing every cited tweet first (C) gave the highest accuracy (0.81), and the two rules in code raised precision to 0.87 (final design).

Table 2. Checker designs on the development topic (McDonalds, 78 claims).

Design	Accuracy	Precision	Recall	False UNKNOWN	False SUPPORTED
Original checker	0.63	0.74	0.38	23	5
A: explicit rules, 32 KB requests, 1,500 answer tokens	0.69	0.64	0.78	8	16
B: A + explanation and quotes before the verdict	0.76	0.82	0.62	14	5
D: model lists missing details, code applies counting rules	0.69	0.88	0.41	22	2
C: B + every cited tweet shown first	0.81	0.84	0.73	9	5
Final: C + two rules in code	**0.82**	**0.87**	**0.73**	**10**	**4**

Precision and recall are for SUPPORTED. The final design was frozen on this topic before the held-out run.

Held-out result

On the held-out topics the revised checker raised accuracy from 0.69 to 0.80, precision from 0.82 to 0.90 and recall from 0.39 to 0.61 (Table 3). False UNKNOWN fell from 34 to 22 and false SUPPORTED from 5 to 4. It corrected 21 verdicts and broke 7 (exact McNemar test, $p = 0.0125$). The gain held on both topics (Obama 0.70 → 0.80, Superbowl 0.68 → 0.79) and was larger on the 93 claims with high-confidence labels (accuracy 0.71 → 0.84). No check failed for technical reasons; one false CONTRADICTED remained.

Table 3. Original and revised checker on the held-out topics (Obama and Superbowl, 132 claims; 57 supported, 75 unknown by the reference).

	Original	Revised
Accuracy	**0.69**	**0.80**
Precision (SUPPORTED)	0.82	0.90
Recall (SUPPORTED)	0.39	0.61
False UNKNOWN	34	22
False SUPPORTED	5	4
False CONTRADICTED	2	1
Accuracy, Obama (76 claims)	0.70	0.80
Accuracy, Superbowl (56 claims)	0.68	0.79
Accuracy, high-confidence labels (93)	0.71	0.84
Precision, high-confidence labels	0.84	0.94
Recall, high-confidence labels	0.49	0.72

Paired on the 132 claims: 21 verdicts corrected, 7 broken; exact McNemar test p = 0.0125.

Results: the final runs

Fact verdicts

With the revised checker, 83 of the 210 draft claims (40%) were SUPPORTED, compared with 46 (22%) before, and none was CONTRADICTED (Table 4). Because the agent revises only contradicted claims, it changed none of the 18 summaries. The earlier finding that the agent reduced contradicted claims from 3 to 0 therefore reflected checker errors, not factual errors in the drafts; the reference labels also contain no contradicted claim. With the proportion layer, one summary changed (Superbowl, prompt 2, seed 202), adding one SUPPORTED claim.

Table 4. Fact verdicts of the summary claims.

Condition	SUPPORTED	CONTRADICTED	UNKNOWN	Claims
Drafts, original checker	46 (22%)	3 (1%)	161 (77%)	210
After the agent, original checker (poster)	50	0	162	212
Drafts, revised checker	**83 (40%)**	**0**	**127 (60%)**	**210**
After the agent, revised checker	83	0	127	210
After the agent, revised checker + proportion layer	84	0	127	211

The number of claims changes when a summary is revised.

Fairness

Mean Equal Coverage was 0.0024 in all four conditions (Table 5): neither the agent nor the proportion layer changed it. The drafts covered the three groups unequally, but not consistently against one group (Table 6). Mean coverage was 2.20% for AA, 2.59% for Hispanic and 2.39% for White tweets; the worst-covered group was AA in 7 summaries, White in 8 and Hispanic in 3; and the differences were not significant (Friedman test, p = 0.35; AA vs Hispanic, Wilcoxon signed-rank test, p = 0.10). Still, 13 of the 18 summaries failed the four-fifths rule: on average the worst-covered group received 73% of the coverage of the best-covered group. Per

topic (Table 7), McDonalds had the lowest EC (0.0013) but the highest disparity (0.34): its coverage is low overall, so the absolute gaps that EC measures are small while the relative gap is the largest. In the run with the original checker, the three revisions triggered by false contradictions happened to raise the worst-covered group in each of those summaries (mean worst-group coverage 2.07% → 2.17%), but because they were triggered by checker errors this is not an effect of the design.

Table 5. Fairness of the 18 summaries per condition (means; labels used only here).

	Drafts (no agent)	Agent, original checker (poster)	Agent, revised checker	Revised checker + proportion layer
Summaries changed	–	3/18	0/18	1/18
Equal Coverage (lower is fairer)	**0.0024**	**0.0024**	**0.0024**	**0.0024**
Worst-group coverage	2.07%	2.17%	2.07%	2.02%
Four-fifths disparity	0.267	0.260	0.267	0.271
Summaries passing the four-fifths rule (≤ 0.20)	5/18	6/18	5/18	5/18

Table 6. Coverage per dialect group in the 18 drafts.

Group	Mean coverage	Worst-covered group in
AA	2.20%	7 summaries
Hispanic	2.59%	3 summaries
White	2.39%	8 summaries

Friedman test over the 18 drafts: p = 0.35. Wilcoxon signed-rank: AA vs Hispanic p = 0.10, AA vs White p = 0.21, Hispanic vs White p = 0.18.

Table 7. Fairness of the drafts per topic (6 summaries each).

Topic	EC	Worst group	Four-fifths disparity	AA	Hispanic	White	Worst-covered most often
McDonalds	0.0013	0.59%	0.343	0.89%	0.81%	0.64%	White (5 of 6)
Obama	0.0035	3.07%	0.241	3.12%	4.05%	3.51%	AA (4 of 6)
Superbowl	0.0025	2.54%	0.218	2.60%	2.90%	3.02%	AA (3 of 6)
All 18	**0.0024**	**2.07%**	**0.267**	**2.20%**	**2.59%**	**2.39%**	**White (8), AA (7)**

Proportion feedback layer

The proportion layer judged 5 of the 18 drafts proportional (Table 8). In 9 drafts the viewpoint it flagged as under-represented was already reported by a verified claim, so no rewrite was made; 2 feedbacks were invalid; and 2 drafts were judged disproportionate. In Superbowl P1 s303 all three rewrites were rolled back because the checker could not verify them. In Superbowl P2 s202 one rewrite was accepted, but by the labels it made coverage less equal: EC 0.0032 → 0.0033, worst-covered group 2.76% → 1.90%, four-fifths disparity 0.22 → 0.29.

Table 8. Outcomes of the proportion feedback layer (18 drafts).

Outcome	Drafts	Result
Judged proportional	5	no change
Flagged viewpoint already reported by a verified claim	9	no change
Invalid feedback	2	no change
Disproportionate: Superbowl P1 s303	1	3 rewrites rolled back (not verified)
Disproportionate: Superbowl P2 s202	1	1 rewrite accepted; EC 0.0032 → 0.0033

Discussion

Discussion

Validating the fact checker changed our conclusions. The original checker was too strict to confirm most supported claims, and its few contradictions were errors; those errors were the only revisions the agent made. With a checker whose accuracy holds on held-out topics, Qwen's drafts contain no contradicted claims, so the self-correction loop has nothing to correct. Neither feedback loop changed fairness: Equal Coverage stayed at 0.0024 in every condition. The drafts do show coverage gaps between dialect groups that fail the four-fifths rule in most summaries, but the disadvantaged group varies between summaries and topics, and no group is significantly under-covered. By design the agent never sees dialect groups, which plausibly explains why it cannot target these gaps; the one rewrite of the proportion layer made coverage less equal.

Most remaining UNKNOWN verdicts come from wrong citations rather than from the checker: half of the claims not supported by their cited tweets are supported by another tweet of the topic. Citation repair, which ranks the topic's uncited tweets by embedding similarity and idf-weighted word overlap, puts a supporting tweet in the top five for 38 of these 58 claims and is the most promising next step.

Limitations

Limitations

The reference labels come from one annotator, an LLM; the agreement of two human annotators on a sample of 60 claims (`κ = [..]`) qualifies every checker result. Recall of SUPPORTED on the held-out topics is 0.61, so about four in ten supported claims are still not confirmed, and one false contradiction remains; we did not widen the rule after seeing it, to keep the test unbiased. Qwen is not reproducible across sessions at temperature 0: all results come from single runs, and the McNemar comparison includes this run-to-run variation because the original verdicts come from an earlier session. With 18 summaries the tests of group differences have little power, so the absence of a significant difference is not evidence of equal coverage. Coverage is measured with MNLI entailment probabilities, which are small (around 2%) because most pairs of a tweet and a summary sentence are unrelated, and EC, an absolute measure, depends on that overall level. Finally, citation repair is not implemented, so wrongly cited claims remain UNKNOWN.

Open task: human labels for κ

Two team members each label the same 60 claims in `annotation_sheet_60.csv` (Saif sends the file), alone and without discussing until both are done. Fill the column `your_verdict` and save the file as `annotation_sheet_60_NAME.csv` . Saif then runs `scripts/kappa.py` and fills in the three κ values.

Labelling rules: use only the cited tweets

1. SUPPORTED when a cited tweet shows the same content; paraphrase counts. Motives, causes or judgments the tweet does not express do not count.
2. "some", "others", a plural ("users", "people") or no count word: 1 supporting tweet is enough.
3. "several", "multiple", "many", "numerous", "frequently", "often", "recurring", "common", "a number of": 3 different supporting tweets.
4. "most", "majority", "a significant or large portion/number", "a minority", "generally", "overall": always UNKNOWN.
5. "mixed", "diverse", "a range of reactions": at least 2 clearly different reactions (3 for "wide range" or "variety").
6. Lists (A, B and C): every item needs support.
7. Other tweets saying something else never count against "some users say X".
8. CONTRADICTED only when a cited tweet explicitly says the opposite of the claim itself. Missing content is UNKNOWN.

Where the data is

- `results/agent_checker_v2` : main run, 18 trials and `evaluation.json` .
- `results/agent_checker_v2_proportion` : run with the proportion layer.
- `results/checker_evaluation` : reference labels, guideline, checker runs, scripts, annotation sheet.
- `docs/CHECKER_EVALUATION.md` : the full write-up of the checker revision.

These files are on Saif's laptop and not yet committed to the repository.